Fiscal Year:	FY 2008	**Task Last Updated:**	FY 12/16/2008
PI Name:	Cavanagh, Peter R. Ph.D., D.Sc.		
Project Title:	Foot Reaction Forces During Simulated ISS Exercise Countermeasures		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Bone Loss Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	98195-6500	**Congressional District:**	7
Comments:	PI moved from Cleveland Clinic to University of Washington in June 2008 (8/08)		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	02/01/2004	**End Date:**	04/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 4/30/2008 per K. Major/NSBRI (3/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Perusek, Gail (NASA GRC)		
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Performance Goal No.:			
Performance Goal Text:			

AIMS: The original aims of this project were: (1) To improve the biomechanical validity of the human supine suspension approach to microgravity simulation and to develop hardware and software that will characterize the force magnitude (F), rate of change of force (dF/dt), and vibratory and transmission characteristics of load transferred to the human body. (2) To fully characterize lower-extremity loading during high-fidelity simulated 0g exercise on various exercise devices throughout the range of SLD and harness settings. (3) To propose a regimen of exercise on the available devices that will be effective in maintaining bone and muscle mass.
KEY FINDINGS: The main findings and deliverables from this study are:

1. HARNESS AND SUBJECT LOAD DEVICE: Locomotor exercise in reduced gravity requires a harness attached to a

Task Description:	subject load device to return the subject to the treadmill surface. The project team and its collaborators have developed enhanced versions of both these devices. A new pneumatic Subject Load Device (pSLD) has been shown to reduce variability in the applied load to within 7% at full bodyweight loading. This is superior to all other forms of SLD that we have previously examined. It has also been shown to be reliable in more than 100 hours of subject testing. The Cleveland Clinic (CC) prototype harness for use with exercise countermeasures has been developed and fabricated. Visual analog scale (VAS) comparisons with a replica of the NASA TVIS harness during short duration trials indicated less discomfort in the shoulders using the CC harness. Overall discomfort ratings were less than or equal to the TVIS harness depending on the loading conditions. During long duration trials the relative overall discomfort was less with the CC harness than the TVIS harness. The success of this development and the harness comfort studies have led to the advancement of this device towards a Detailed Science Objective (DSO) which is currently scheduled for a flight experiment by our colleagues at NASA GRC and ZIN technologies. 2. THE ENHANCED DAILY LOAD STIMULUS MODEL: The project team has modified existing mathematical theories on daily load stimulus (DLS) to develop a comprehensive algorithm, accounting for new theories on saturation and recovery of the osteogenic potential of bone with distributed mechanical loading and the stimulus provided to the bone through standing. We believe that the new algorithm, the enhanced DLS (eDLS), will provide a better model of the relationship of bone mineral density to daily loading and are currently testing it as a method of prescribing an exercise “dose” to subjects in an ongoing bedrest study. We have validated this model against bed-rest data and have submitted it to the journal “Aviation, Space and Environmental Medicine”. 3. STAND-ALONE ZLS (sZLS): Our collaborators at Glenn Research Center have leveraged their experience gained on the development of the eZLS into a new device, the Standalone Zero Gravity Locomotion Simulator (sZLS) which has been delivered to the NASA Flight Analogs Facility at the University of Texas Medical Branch in Galveston, TX. 4. GROUND REACTION FORCES WITH A COMPLIANT INTERFACE: A computational model of the eZLS environment was created in Simulink to understand system dynamics and their effect on foot reaction forces between the crewmember and the exercise countermeasure device. The model simulated rigid body dynamics of the treadmill and its supporting frame or “rack” as the simulated crewmember foot forces excite the dynamics of the isolator components. The model predicted that with a vibration isolation system installed on the treadmill the peak reaction force on the foot could drop by up to 14% compared to a grounded system. This finding is significant because: A) astronaut crew exercise prescriptions are formulated based on a grounded system; and B) the on-orbit exercise systems used by astronauts are usually equipped with a vibration isolation system. The implication is that astronauts are not stimulating the musculoskeletal system as much as desired and we believe this could be a contributing factor in the persistence of bone loss on long duration space missions despite rigorous exercise. In a subsequent experiment with human subjects, the measured peak ground reaction forces during running were not found to show a major dependence on treadmill compliance within the bounds of the compliance variation used in the study. This finding is unexpected and does not agree with the results of the simulation. The authors are currently refining the model to consider an energy analysis similar to that used in shock analysis. 5. NEW EXERCISE DEVICES: Two new small footprint, low power exercise devices have been designed, fabricated, and tested. Both of these devices (the Impacting Cycle Ergometer [ICE] and the Impacting Stepper Device [ISD]) provide a calibrated impact to the foot in order to enhance the loading profile. Based on subject feedback, measured heart rate, and foot force measurement, the ISD appears worth further investigation. IMPACT: These finding provide new opportunities for optimizing in-flight exercise by providing: 1) a harness and subject load device that will allow greater, and more consistent, loading than was previously possible; 2) the enhanced daily load stimulus model for more accurate matching of individual 1G cumulative loads with mission exercise loads; 3) Experimental testbeds (the eZLS and sZLS) that are available for accurate ground based simulations of exercise in altered gravity environments; 4) initial insights in the model-predicted reductions that may occur on exercise devices with compliant interfaces; 5) New small footprint, low power exercise devices for use in long-duration missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Knowledge gained from this research will contribute to a better understanding of the importance of exercise for the development and maintenance of bone strength among humans living in Earth gravitational fields. Knowledge in this area is crucial to the treatment of bone disease for which exercise may or may not be an effective intervention.
Task Progress:	1. eZLS. Mechanical Design and Fabrication. Construction and system integration activities for the eZLS are complete. This facility is now in continuous use at NASA GRC. Our collaborators at NASA GRC have leveraged their experience gained on the development of the eZLS into a new device, the Stand-alone Zero Gravity Locomotion Simulator (sZLS) which has now been delivered to the NASA Flight Analogs Facility at UTMB in Galveston, TX. 2. pSLD. NASA GRC and CCF collaborated to create a passive pneumatic subject load device (pSLD). This device uses a single pneumatic cylinder to provide a nearly constant spring force on the cables attached to the subject harness. This system has been successfully installed at the Cleveland Clinic and has been used for human subject testing at the Cleveland Clinic. 3. Harness Design. The initial design, development, and evaluation of the new harness has been completed. The harness has been successfully used in the experiments described below on the new exercise devices and for a related bedrest study. An evaluation of the new harness by two ISS crewmembers was also conducted. This harness has been well received and the device is now scheduled for flight on the ISS in a Detailed Science Objective (DSO) led by our colleagues at NASA GRC and ZIN technologies. 4. Experiments on Compliant Interfaces. These experiments have been completed. The measured peak ground reaction forces during running were not found to show a major dependence on treadmill compliance within the bounds of the compliance variation used in this study. This finding is unexpected and does not agree with the results of the simulation conducted earlier in the project. The authors are currently refining the model to consider an energy analysis similar to that used in shock analysis. 5. Exercise Prescription. The modification of the Daily Load Stimulus (DLS) model has been completed. The revised algorithm, the enhanced DLS (eDLS), incorporated both magnitude and time domain effects (cyclic nature of typical

	loading). The revised algorithm has been implemented into a toolkit for the processing of test subject ground reaction force data and a manuscript has been submitted to Aviation, Space, and Environmental Medicine.
Bibliography Type:	Description: (Last Updated: 03/08/2018)
Articles in Peer-reviewed Journals	Genc KO, Humphreys BT, Rice AJ, Englehaupt RK, Novotny SC, Gopalakrishnan R, Ilaslan H, Licata AA, Cavanagh PR. "Enhanced daily load stimulus: accounting for cyclical loading and standing on the osteogenic response of bone." Aviat Space Environ Med. Submitted, July 2008. , Jul-2008
Awards	Cavanagh PR. "Laurence R. Young Space Biomedical Research Award, NASA/NSBRI International Academy of Astronautics, Elected as Corresponding Member, September 2007." Sep-2007
Books/Book Chapters	Cavanagh PR, Rice AJ, Licata AA. "40 years of bone loss in space." in "Bone loss during spaceflight : etiology, countermeasures, and implications for bone health on Earth." Ed. P.R. Cavanagh, A.J. Rice. Cleveland : Cleveland Clinic Press, c2007. p. 1-15, Apr-2007
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